

Intro to Data Science, Engineering, and AI

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2026

SIT, KMUTT

INT182: Data Science and Artificial Intelligence Fundamentals

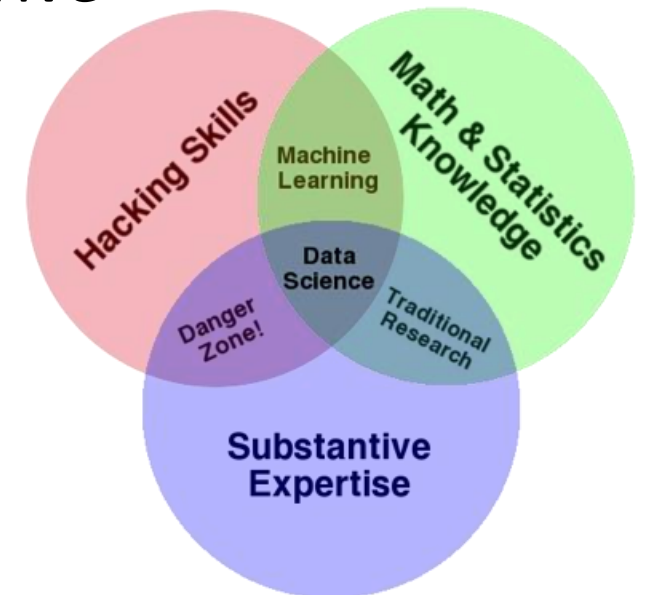
Introduction to data science, data collection and preprocessing, exploratory data analysis, feature engineering, introduction to machine learning, basic model evaluation, predictive analysis, data science tools, applications and impact of data science, ethical considerations in data science, Introduction to artificial intelligence, search algorithms and optimization, knowledge representation and reasoning, introduction to deep learning, artificial intelligence in real-world applications, artificial intelligence tools and platforms, ethical considerations in artificial intelligence, future trends and emerging technologies in artificial intelligence

ผลลัพธ์การเรียนรู้ระดับรายวิชา:

- (1-I; 2B-I; 2C-I; 2D-I; 5AI-I; 5DS-I; 5DA-I) Describe and explain the fundamental concepts, techniques, and ethical considerations in data science and artificial intelligence, and analyze their impact on industries, society, and everyday life in diverse real-world scenarios.

What Data Science (DS) is all about...

- **Data science** is the interdisciplinary field that involves extracting meaningful insights and knowledge from large volumes of data.
- It's a powerful tool used across various industries to make informed decisions, optimize processes, and drive innovation.
- **In essence, data science is about turning data into actionable insights that drive business value.** It's a rapidly evolving field with immense potential for innovation and problem-solving.



Data Engineering

Data engineering is the practice of designing, building, and maintaining systems and infrastructure to collect, store, process, and analyze large volumes of data. It focuses on creating reliable, scalable pipelines to transform raw data into formats suitable for analysis, machine learning, or business intelligence. Key tasks include:

- Data Collection: Extracting data from various sources like databases, APIs, or logs.
- Data Storage: Managing data in databases, data warehouses, or data lakes (e.g., SQL, NoSQL, Hadoop, Snowflake).
- Data Processing: Cleaning, transforming, and enriching data using tools like Apache Spark, Airflow, or Kafka.
- Pipeline Automation: Building ETL (Extract, Transform, Load) or ELT pipelines to automate data workflows.
- Data Quality: Ensuring data accuracy, consistency, and accessibility.
- Scalability & Performance: Optimizing systems to handle large-scale data efficiently.

Data engineers work closely with data scientists and analysts, enabling them to access high-quality data for insights or model training. It's a blend of software engineering, database management, and cloud technologies, often involving programming (Python, Java, Scala), SQL, and cloud platforms (AWS, Azure, GCP).

Main Components of DS

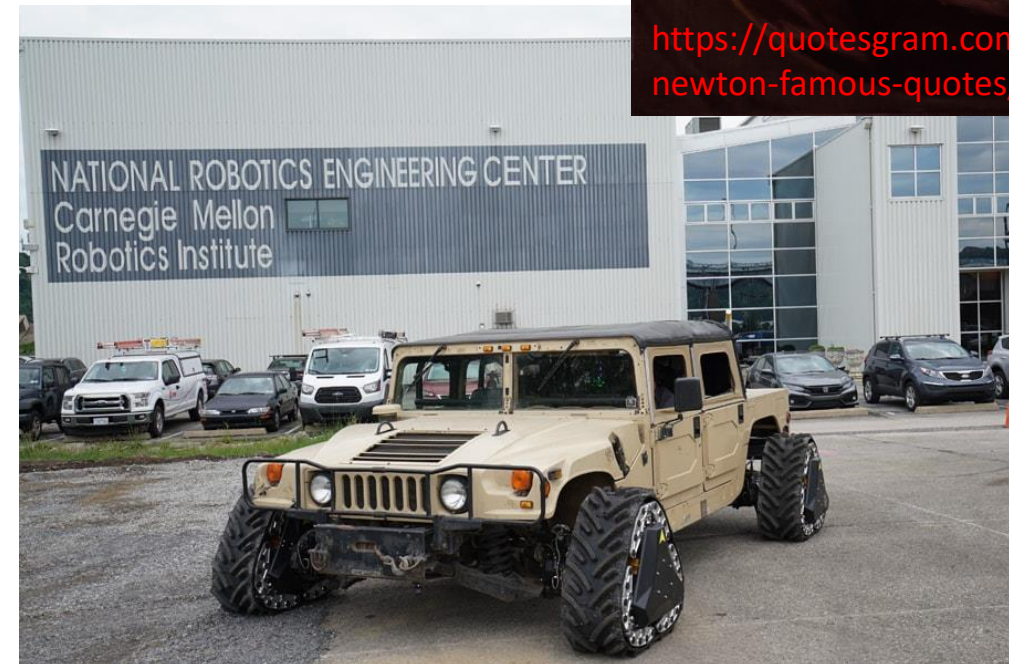
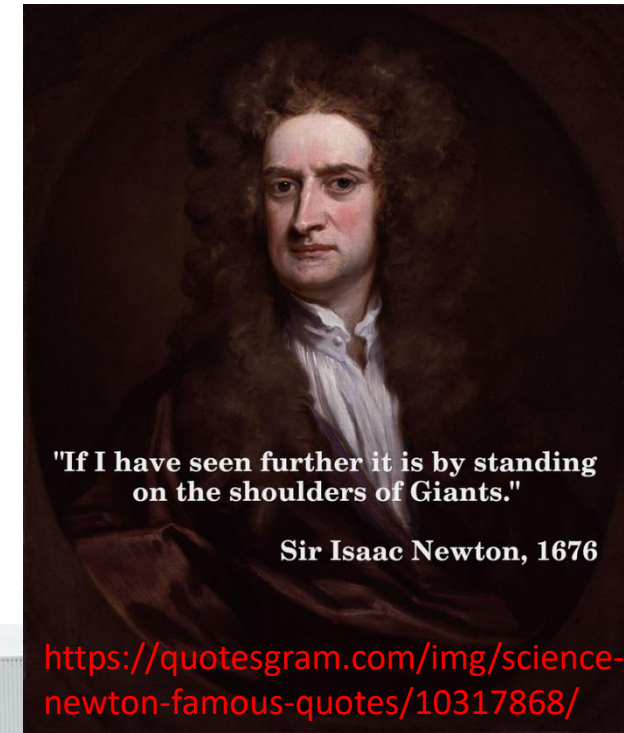
- **Data Collection:** Gathering data from diverse sources, including databases, spreadsheets, social media, sensors, and more.
- **Data Cleaning and Preparation:** Transforming raw data into a suitable format for analysis, handling missing values, outliers, and inconsistencies.
- **Exploratory Data Analysis (EDA):** Summarizing and visualizing data to discover patterns, trends, and anomalies.
- **Data Modeling:** Building statistical or machine learning models to predict future outcomes or classify data points.
- **Evaluation:** Assessing the performance of models and refining them as needed.
- **Data Communication and Visualization:** Presenting findings effectively through visualizations, reports, and presentations.

Key Skills for Data Scientists

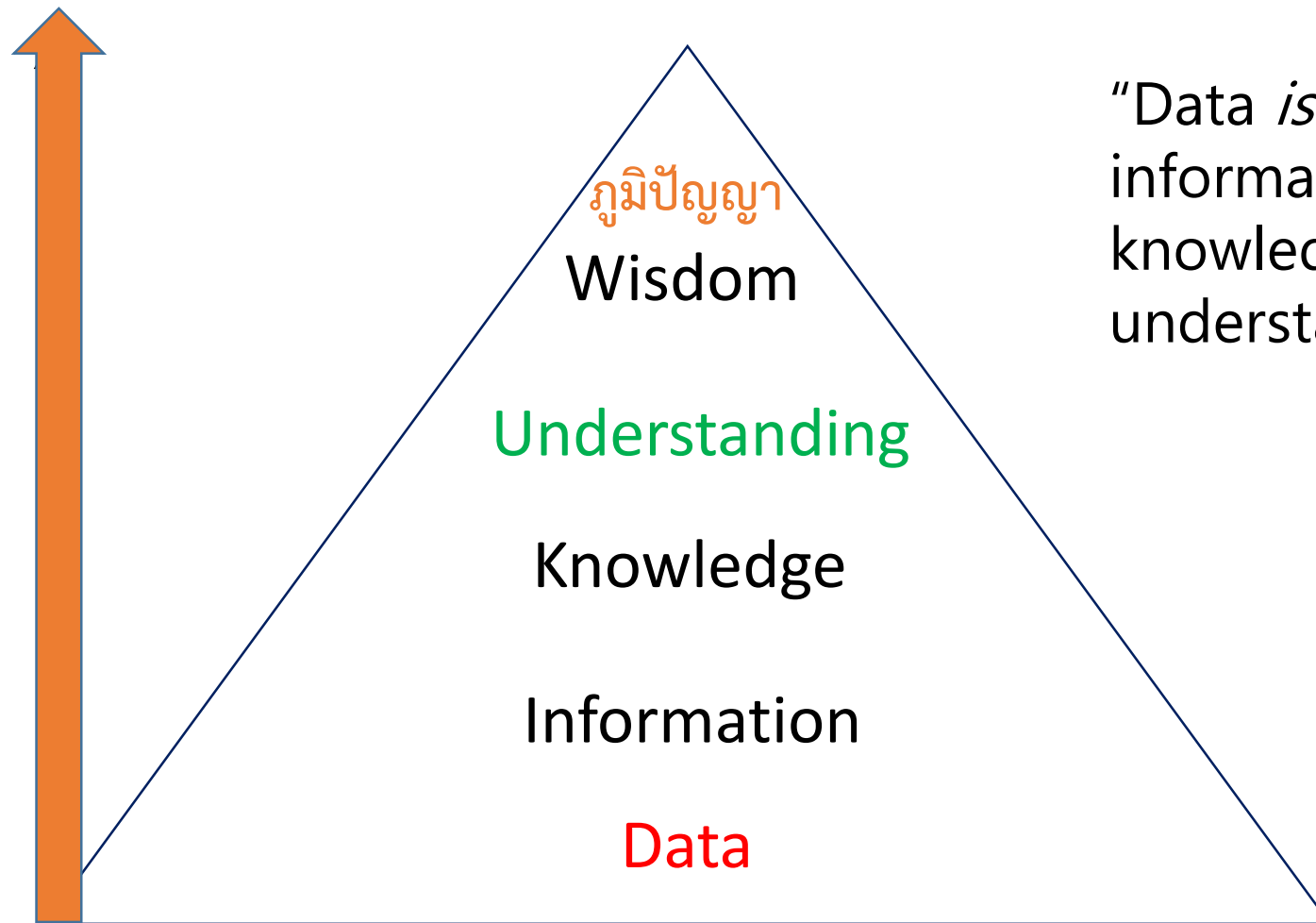
- **Statistical Knowledge:** Understanding probability, hypothesis testing, regression, and other statistical techniques.
- **Programming Proficiency:** Expertise in languages like Python, R, or SQL for data manipulation, analysis, and model development.
- **Machine Learning:** Applying algorithms to learn patterns from data and make predictions.
- **Data Visualization:** Creating clear and informative visualizations to communicate insights.
- **Domain Expertise:** Understanding the specific industry or problem area to apply data science effectively.
- **Critical Thinking and Problem-Solving:** Identifying relevant questions, analyzing data, and drawing meaningful conclusions.

Suggestions of learning DS

- No need of reinventing the wheel
 - ✓ Don't waste time learning how to do something when it is already known how to do it
- Standing on the shoulders of giants
 - ✓ Using the understanding gained by major thinkers who have gone before in order to make intellectual progress
 - ✓ Discovering truth by building on previous discoveries



Data, Information, Knowledge, Understanding, Wisdom



*"Data is not information,
information is not knowledge,
knowledge is not understanding,
understanding is not wisdom."*

-- Clifford Stoll,
American astronomer

To understand the data



“Data don’t make any sense,
we will have to resort to statistics.”

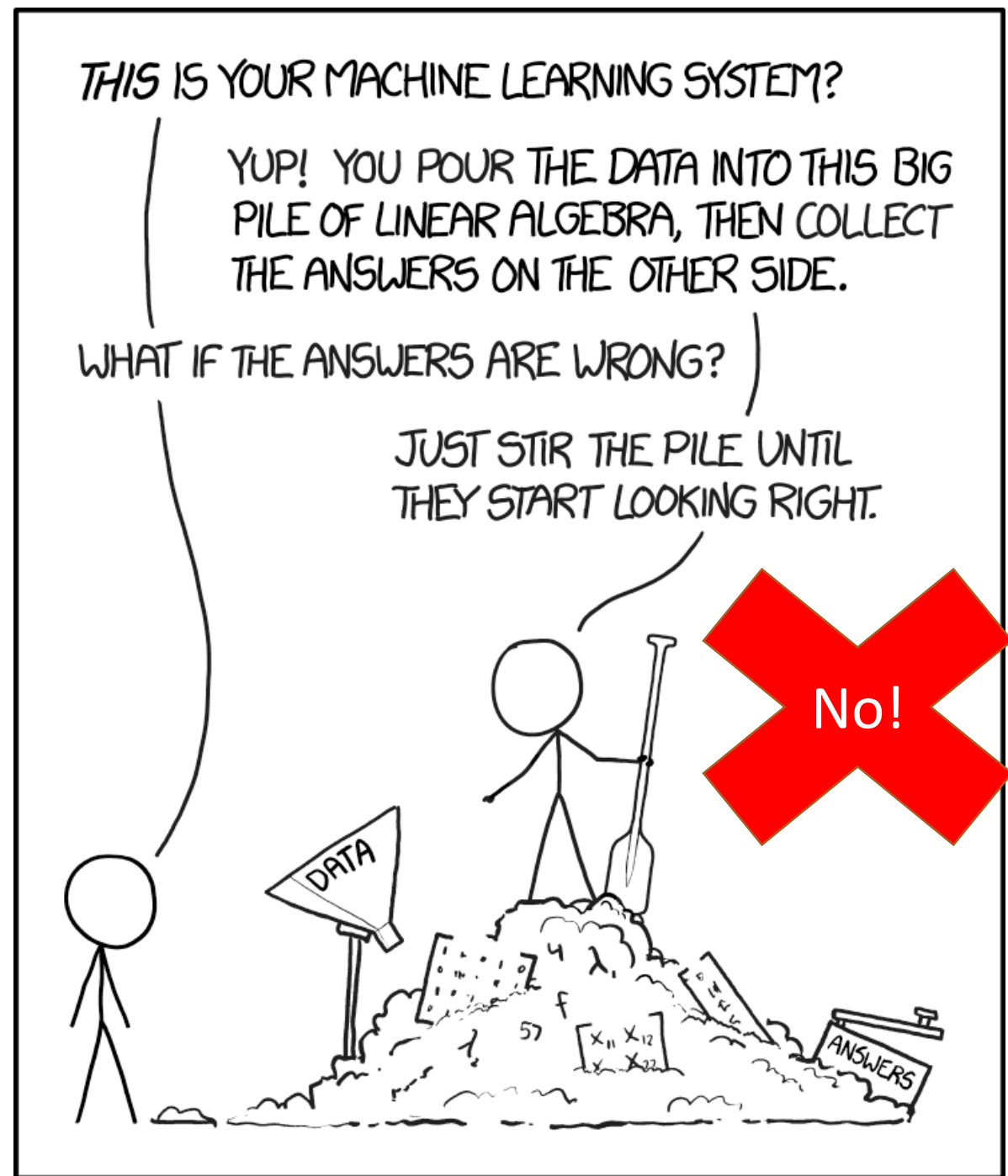
“The most important aspect of a
statistical analysis is not what you do
with the data, it’s what data you use”

-- Andrew Gelman
American statistician

- ✓ Most (almost all) big datasets generated by ML powered systems are biased
- ✓ Bias in data produces biased models which can be discriminatory and harmful to humans
- ✓ A thorough evaluation of the available data and its processing to mitigate biases should be a key step in modeling

We need to find causes and make changes.

Credit: <https://towardsdatascience.com/survey-d4f168791e57>



Data bias

- ✓ Bias when the available data **is not representative** of the population.
- ✓ **Data** includes content produced by humans which **may contain bias** against groups of people.

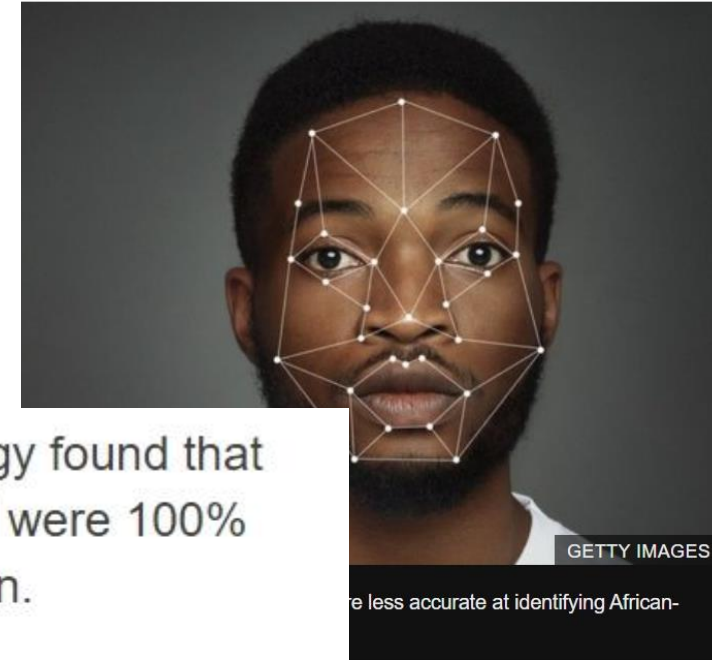
Technology

IBM abandons 'biased' facial recognition tech

🕒 9 June 2020

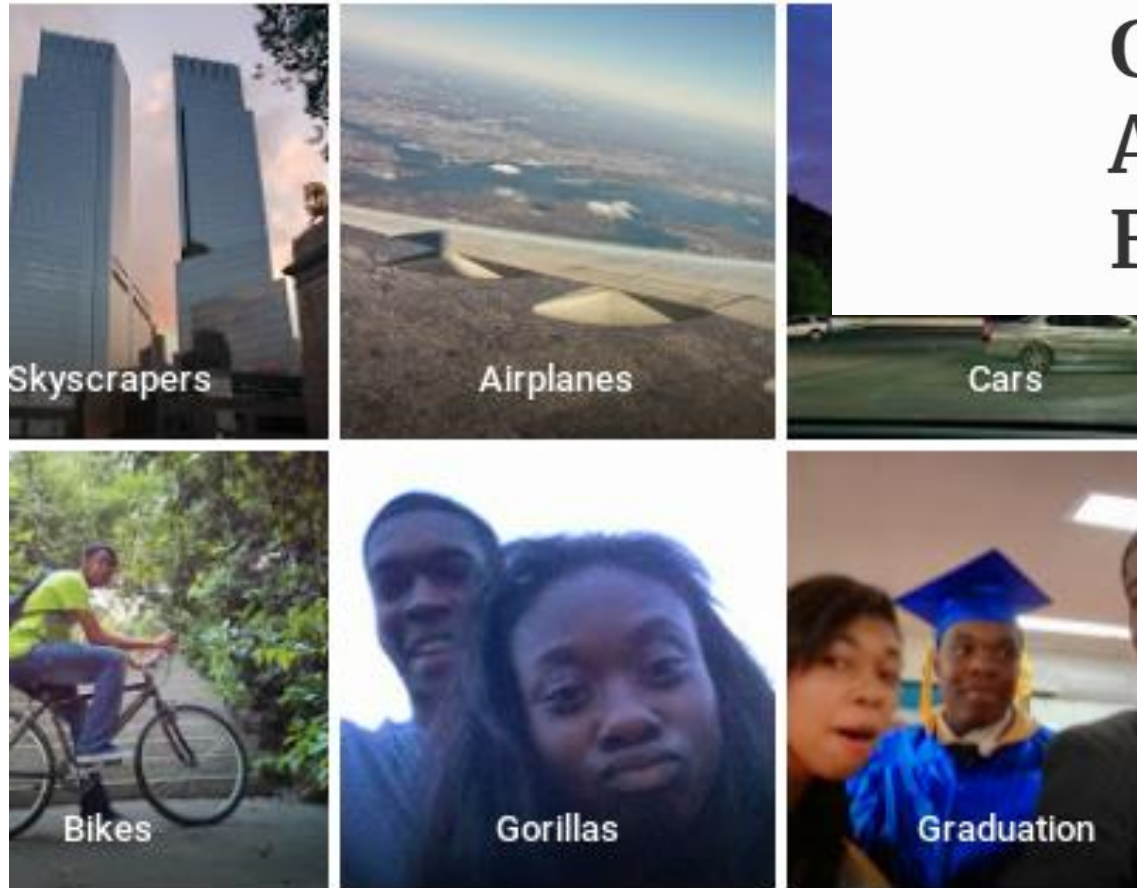
A 2019 study conducted by the Massachusetts Institute of Technology found that none of the facial recognition tools from Microsoft, Amazon and IBM were 100% accurate when it came to recognising men and women with dark skin.

And a study from the US National Institute of Standards and Technology suggested facial recognition algorithms were far less accurate at identifying African-American and Asian faces compared with Caucasian ones.



oftware for "mass

Biased data and not enough training



Forbes

Billionaires Innovation Leadership Money Business Small Business L

Jul 1, 2015, 01:42pm EDT

Google Photos Tags Two African-Americans As Gorillas Through Facial Recognition Software

Credit: <https://www.forbes.com/sites/mzhang/2015/07/01/google-photos-tags-two-african-americans-as-gorillas-through-facial-recognition-software/#6efa9405713d>

Types of biases: Data bias

- **Selection bias**: Occurs when the data used to train the model is not representative of the real-world population.
- **Sampling bias**: A specific type of selection bias where the sample used to train the model is not randomly selected.
- **Confirmation bias**: The tendency to seek out information that confirms existing beliefs, leading to biased data collection.
- **Survivorship bias**: Focusing on successful cases while ignoring failures, leading to a distorted view of reality.
- **Measurement bias**: Errors in data collection or measurement that systematically distort the results.
- **Labeling bias**: Inaccurate or inconsistent labeling of data, affecting the model's learning process

Types of biases: Algorithm bias

- **Algorithmic bias**: Bias inherent in the algorithm itself, often due to design choices or assumptions.
- **Overfitting**: A model that performs well on training data but poorly on new data due to excessive complexity.
- **Underfitting**: A model that is too simple to capture the underlying patterns in the data.

In addition, one example of **human bias** is:

- **Implicit bias**: Unconscious stereotypes or prejudices that influence decision-making.

How to mitigate biases

- **Collect diverse and representative data:** Ensure the data reflects the real world and avoids biases.
- **Use robust algorithms:** Choose algorithms that are less prone to bias and consider fairness metrics.
- **Be aware of human biases:** Recognize and mitigate the impact of human biases in the development process.
- **Monitor and evaluate models:** Continuously assess model performance and identify potential biases.
- **Transparency and explainability:** Understand how models make decisions to identify and address biases

Quote about data

“I have no data yet. It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts.”

-- Sherlock Holmes

Holmes understood the value of combining his data with the art of observation and an awareness of perspective in order to reach elusive insights – a central trait of the data storyteller.

Credit: <https://fuelcommunications.com/sherlock-holmes-a-data-storyteller/>

Holmes is known for his proficiency with observation, deduction, forensic science, and logical reasoning.

https://en.wikipedia.org/wiki/Sherlock_Holmes



<https://www.nytimes.com/2013/03/07/books/su-it-says-sherlock-belongs-to-the-ages.html>

Trust on the data: Transparency and traceability

The evolution of technology is creating a trust crisis. As consumers become more aware of how their data is being collected and used, organizations are also recognizing the increasing liability of storing and gathering the data.

This trend requires a focus on *six key elements of trust*: **Ethics, integrity, openness, accountability, competence and consistency.**

Legislation, like the European Union's General Data Protection Regulation (**GDPR**), is being enacted around the world, driving evolution and laying the ground rules for organizations.

Thailand's Personal Data Protection Act (PDPA) in 2020

- กฎหมายพรบ.คุ้มครองข้อมูลส่วนบุคคล (PDPA)
- http://www.ratchakitcha.soc.go.th/DATA/PDF/2563/E/164/T_0012.PDF

หน้า ๑๒

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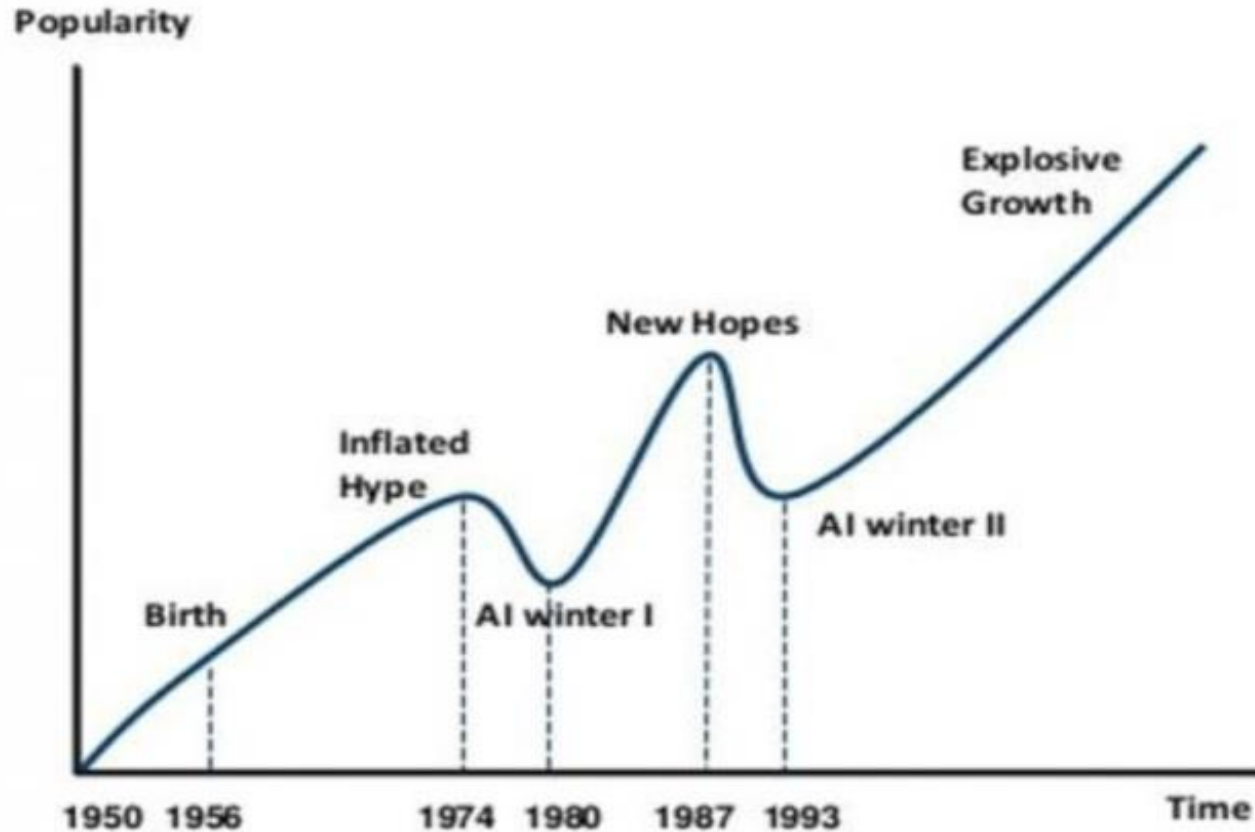
ประกาศกระทรวงดิจิทัลเพื่อเศรษฐกิจและสังคม

เรื่อง มาตรฐานการรักษาความมั่นคงปลอดภัยของข้อมูลส่วนบุคคล พ.ศ. ๒๕๖๓

โดยที่มาตรา ๓ วรรคสอง แห่งพระราชกฤษฎีกากำหนดหน่วยงานและกิจการที่ผู้ควบคุมข้อมูลส่วนบุคคลไม่อยู่ภายใต้บังคับแห่งพระราชบัญญัติคุ้มครองข้อมูลส่วนบุคคล พ.ศ. ๒๕๖๒ พ.ศ. ๒๕๖๓ กำหนดให้ผู้ควบคุมข้อมูลซึ่งเป็นหน่วยงานหรือกิจการตามบัญชีท้ายพระราชกฤษฎีกาดังกล่าวต้องจัดให้มีมาตรการรักษาความมั่นคงปลอดภัยของข้อมูลส่วนบุคคลให้เป็นไปตามมาตรฐานที่กระทรวงดิจิทัลเพื่อเศรษฐกิจและสังคมกำหนด

AI evolution: ups (hype) and downs (winter)

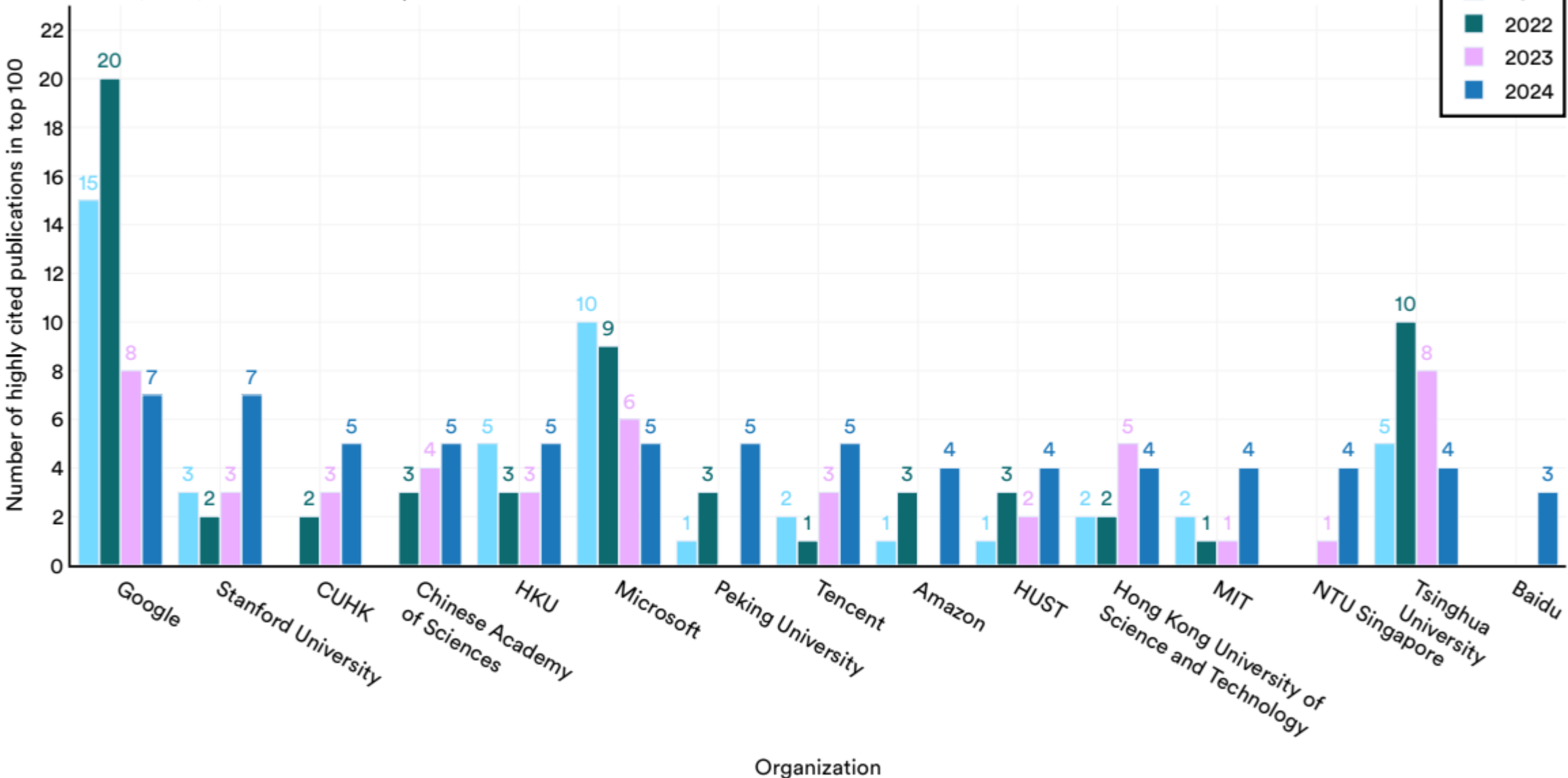
AI HAS A LONG HISTORY OF BEING “THE NEXT BIG THING” ...



Timeline of AI Development	
▪	1950s-1960s: First AI boom - the age of reasoning, prototype AI developed
▪	1970s: AI winter I
▪	1980s-1990s: Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
▪	1990s: AI winter II
▪	1997: Deep Blue beats Gary Kasparov
▪	2006: University of Toronto develops Deep Learning
▪	2011: IBM's Watson won Jeopardy
▪	2016: Go software based on Deep Learning beats world's champions

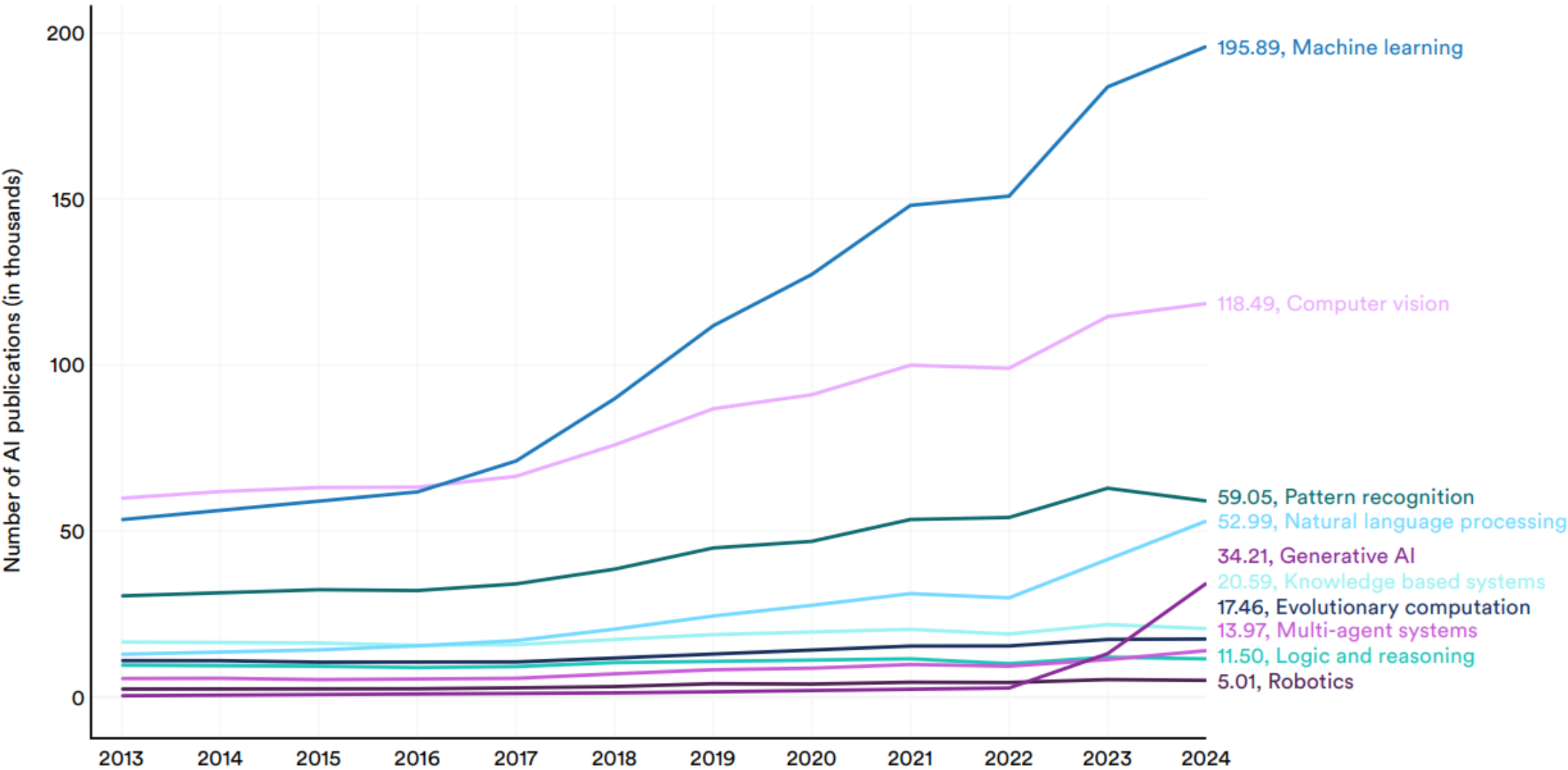
Number of highly cited publications in top 100 by organization, 2021–24

Source: AI Index, 2026 | Chart: 2026 AI Index report



Number of AI publications by select top topics, 2013–24

Source: AI Index, 2026 | Chart: 2026 AI Index report



Scope

World



Years

2016



2026



Adjusted Publications



AI Index

- ▶ AI In General ☒
- ▶ Computer Vision ☒
- ▶ Natural Language ☒
- ▶ Machine Learning ☒
- ▶ Cognitive Reasoning ☒
- ▶ Robotics ☒
- ▶ Multi-Agent System ☒
- ▶ Simulation ☒

Selection Criteria ^①

Institute

City

Country

#		Institute	Adjusted Publications	AI Index
1		Peking University 136 authors ▼	2740.0	169.69
2		Carnegie Mellon University 123 authors ▼	2389.1	160.50
3		Tsinghua University 112 authors ▼	2285.4	136.64
4		Zhejiang University 88 authors ▼	1939.1	82.34
5		Stanford University 53 authors ▼	1704.0	98.82
6		Massachusetts Institute of Technology 65 authors ▼	1684.7	114.28
7		Chinese Academy of Sciences 55 authors ▼	1644.4	68.78
8		Univ. of California - Berkeley 64 authors ▼	1584.7	68.15
9		Nanyang Technological University 49 authors ▼	1476.4	75.65
10		KAIST 62 authors ▼	1453.5	62.34

AI Rankings

How much data is generated daily?

In 2025, there were about **400 quintillion (or 400,000 quadrillion) bytes of new data** being generated around the globe every single day—and that number is still climbing as AI adoption accelerates.

It is enough data every single day to fill roughly 95 to 120 BILLION standard DVDs (4.7 GB per disc)

-- Credit: DeepSeek, July 2026

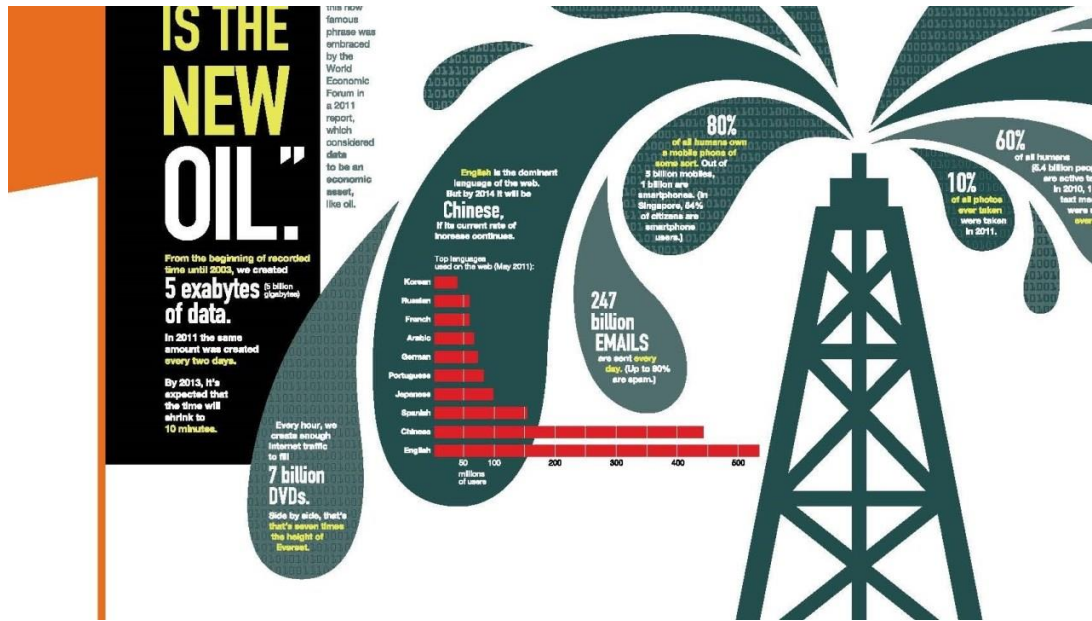
Exabyte = 1 quintillion bytes

Quintillion

= 10^{18}

= million million millions

Data is the **new oil** (C. Humby, 2006)



“Data is the new oil. It’s valuable, but if unrefined it cannot really be used. It has to be changed into gas, plastic, chemicals, etc to create a valuable entity that drives profitable activity; so must data be broken down, analyzed for it to have value.”
-- Clive Humby, 2006

“Data is the new oil,” coined in 2006 by Clive Humby, a British data commercialization entrepreneur, this now famous phrase was embraced by the World Economic Forum in a 2011 report

AI is the **new electricity** (A. Ng, 2016)

Artificial Intelligence is the New Electricity

-- Andrew Ng, former Baidu Chief Scientist, Coursera co-founder, and Stanford Adjunct Professor



About a century go, we started to electrify the world through the electrical revolution. By replacing steam powered machines with those using electricity, we transformed transportation, manufacturing, agriculture, healthcare and so on. Now, AI is poised to start an equally large transformation on many industries.

“Anything that a typical human can do with at most 1 sec of thought, can probably now or soon be automated with AI.” This is an imperfect rule but quite helpful.

Note: 5G latency is about single-digit milliseconds at best

Credit: <https://medium.com/syncedreview/artificial-intelligence-is-the-new-electricity-andrew-ng-cc132ea6264>

Data Science (DS)

- “Data Science refers to an emerging area of work concerned with the collection, preparation, analysis, visualization, management and preservation of large collections of information.”

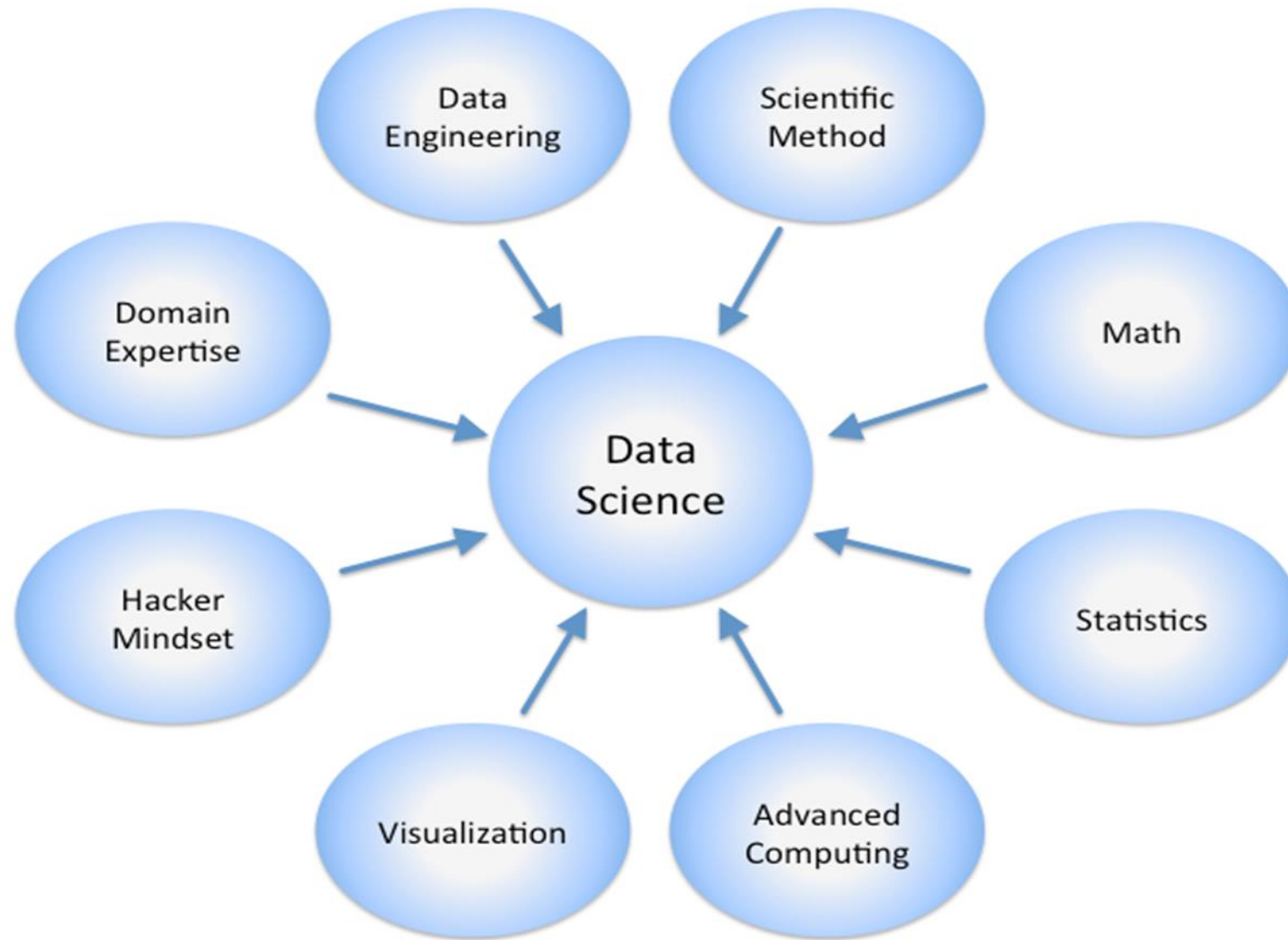
-- Jeffrey Stanton, Syracuse University

- “A data scientist is someone who can obtain, scrub, explore, model and interpret data, blending hacking, statistics and machine learning. Data scientists not only are adept at working with data, but appreciate data itself as a first-class product.”

-- Hilary Mason, chief scientist at bit.ly

The term “**data science**” has been traced back to 1974, when Peter Naur proposed it as an alternative name for computer science.

“Data scientist” has been coined by DJ Patil and Jeff Hammerbacher in 2008



Practical definition of DS

- Data Science is about the whole processing pipeline to extract information out of data. Data Scientists understand and care about the whole data pipeline
- A **data pipeline** consists of 3 steps:
 - 1) Preparing to run a model
 - Gathering, cleaning, integrating, restructuring, transforming, loading, filtering, deleting, combining, merging, verifying, extracting, shaping
 - 2) Running the model
 - 3) Communicating the results

What DS does

- An area that manages, manipulates, extracts, and interprets knowledge from tremendous amount of data
- Data science (DS) is a multidisciplinary field of study with goal to address the challenges in big data
- Data science principles apply to all data – especially big data

Big data: The 5 V's + 2 more V's

- Raw data, big data, unstructured and structured: **Volume**; probably the best known characteristic of big data
- Change over time, speed of incoming data and processing: **Velocity**
- Data types, from many sources: **Variety**
- Data quality, clean & accurate: **Veracity**; confidence or trust in the data
- Information for decision making, how data is used and/or handled: **Value**; arguably the most important of all

+ 2 more V's: **Variability** and **Visualization**

- **Variability**: variability in meaning, be able to understand words in contexts; or the number of inconsistencies in the data; it also refers to the inconsistent speed at which big data is loaded into your database.
- **Visualization**: making the collected and analyzed data understandable and easy to read

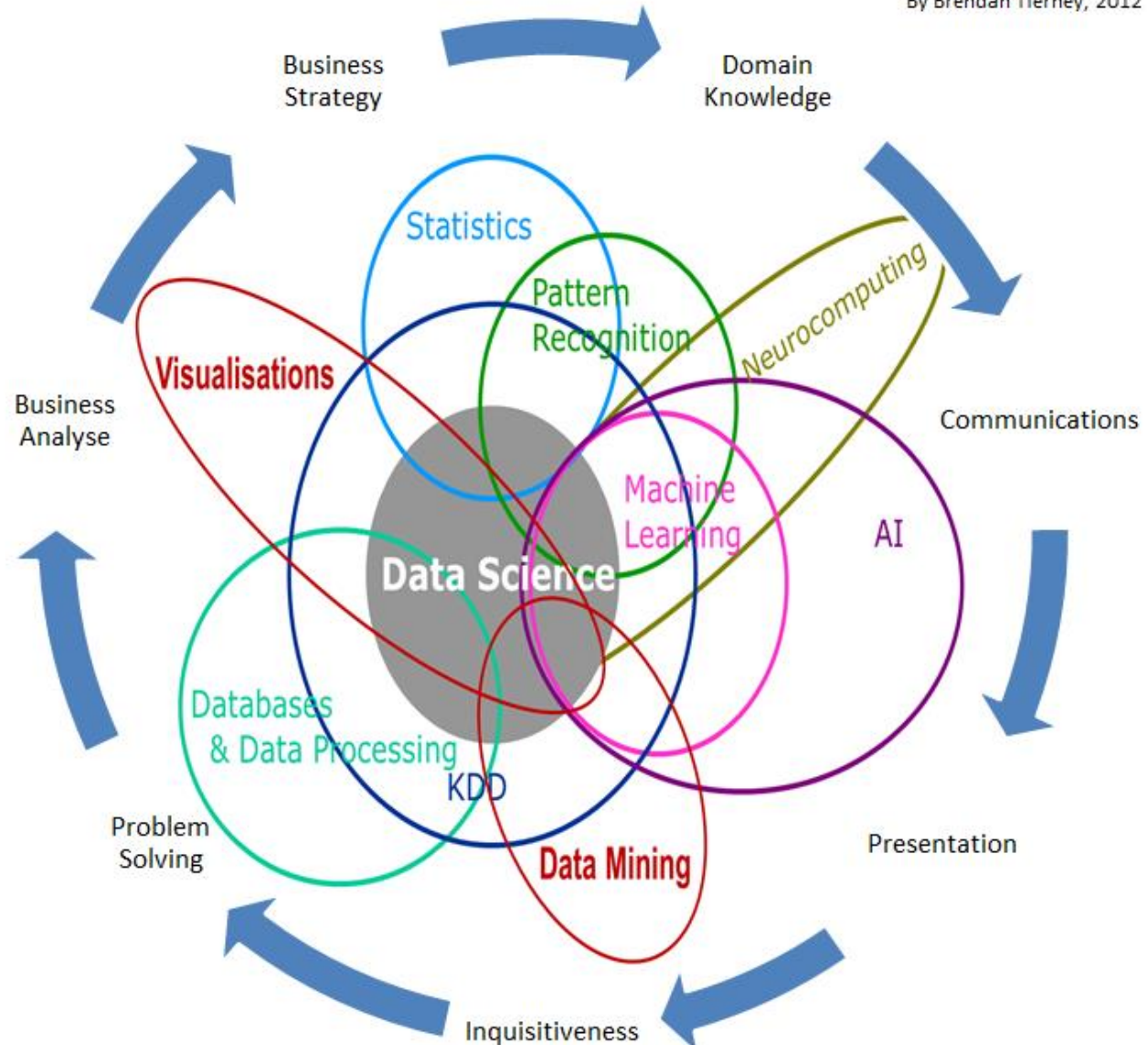
Data Science Is Multidisciplinary

By Brendan Tierney, 2012

DS = “A jack of all trades and a master of some”

“A jack of all trades” means a person who can do many different types of work.

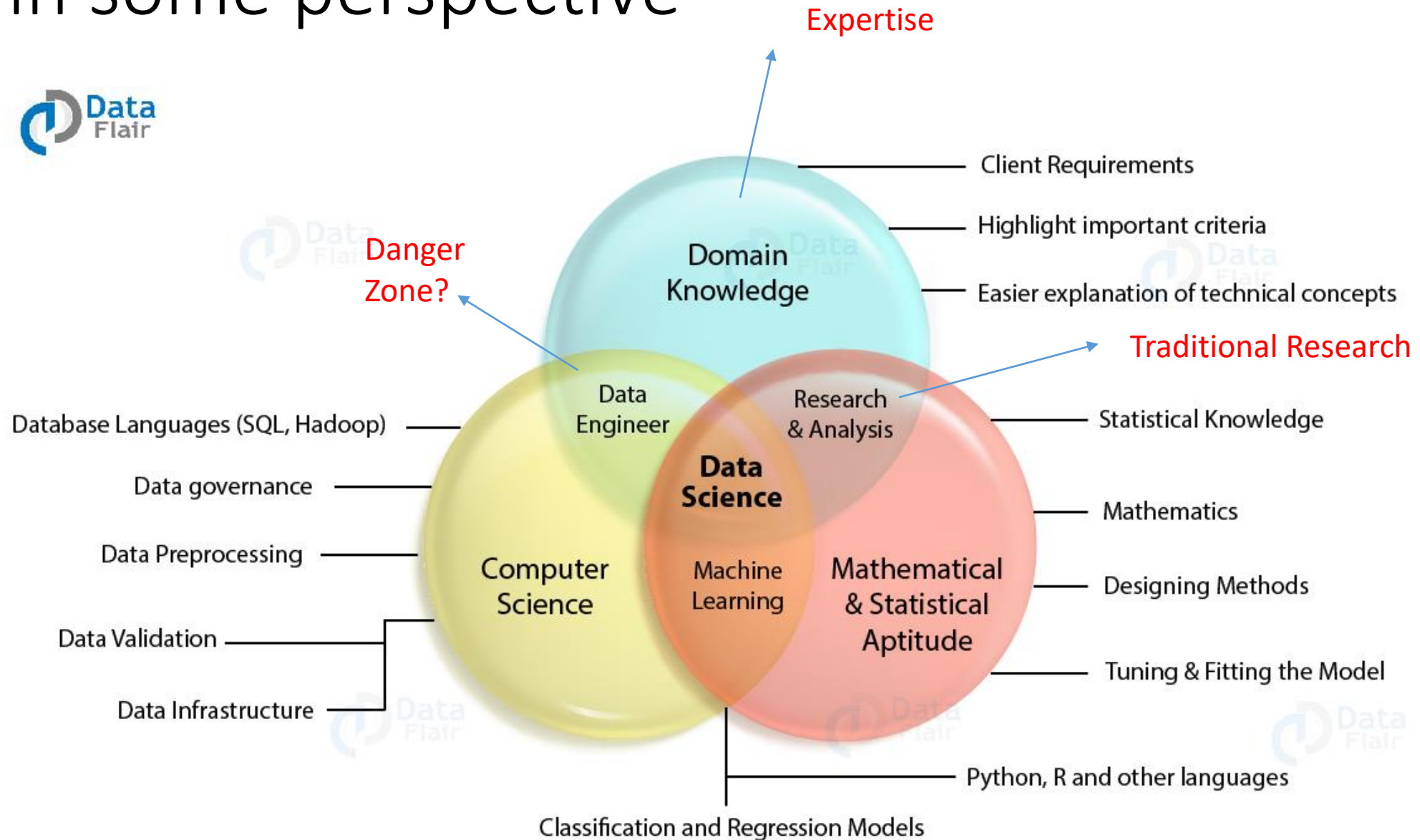
Credit: Brendan Tierney, 2012



What DS needs

- Theories and techniques from many fields and disciplines are used to investigate and analyze a large amount of data to help decision makers in many industries such as science, engineering, economics, politics, finance, and education
 - Computer Science
 - Pattern recognition, visualization, data warehousing, high performance computing, databases, AI
 - Mathematics
 - Mathematical Modeling
 - Statistics
 - Statistical and stochastic modeling, probability.

DS in some perspective



So data science is/does ...

Data Science deals with both structured and unstructured data.

- It is a field that includes everything that is associated with the cleansing, preparation and final analysis of data.
- Data science combines the programming, logical reasoning, mathematics and statistics. It captures data in the most ingenious ways and encourages the ability of looking at things with a different perspective.
- Likewise, it also cleanses, prepares and aligns the data. To put it more simply, data science is an umbrella of several techniques that are used for extracting the information and the insights of data.
- Data scientists are responsible for creating the data products and several other data based applications that deal with data in such a way that conventional systems are unable to do

What the DS is about

Data Science is the science which uses computer science, statistics and machine learning, visualization and human-computer interactions to collect, clean, integrate, analyze, visualize, interact with data to create data products.

- Turn data into information
- Analyze data to get insights
- Identify trends, patterns, and correlations
- Contextualize, apply and understand the results

4 A's of Data Science (DS)

Roles that data scientists play or help others in DS

- Data **A**rchitecture
- Data **A**cquisition
- Data **A**nalysis
- Data **A**rchiving

DS life cycle

1. Requirement engineering
2. Feature engineering (or feature extraction or feature discovery is the process of using domain knowledge to extract features, e.g., characteristics, properties, or attributes/variables from raw data)
3. Feature selection (process of selecting a subset of relevant features)
4. Modelling
5. Deployment and maintenance

Data Science --7 key steps --

Steps taken by DS

1. Business problem

- Ask “why” questions, ทำตัวเป็นแบบเนรน้อยเจ้าปัญญา

2. Data acquisition

3. Data preparation

- Cleaning and transformation
- This is the most time consuming process

4. Exploratory data analysis

- Features and parameters
- This is the most important step

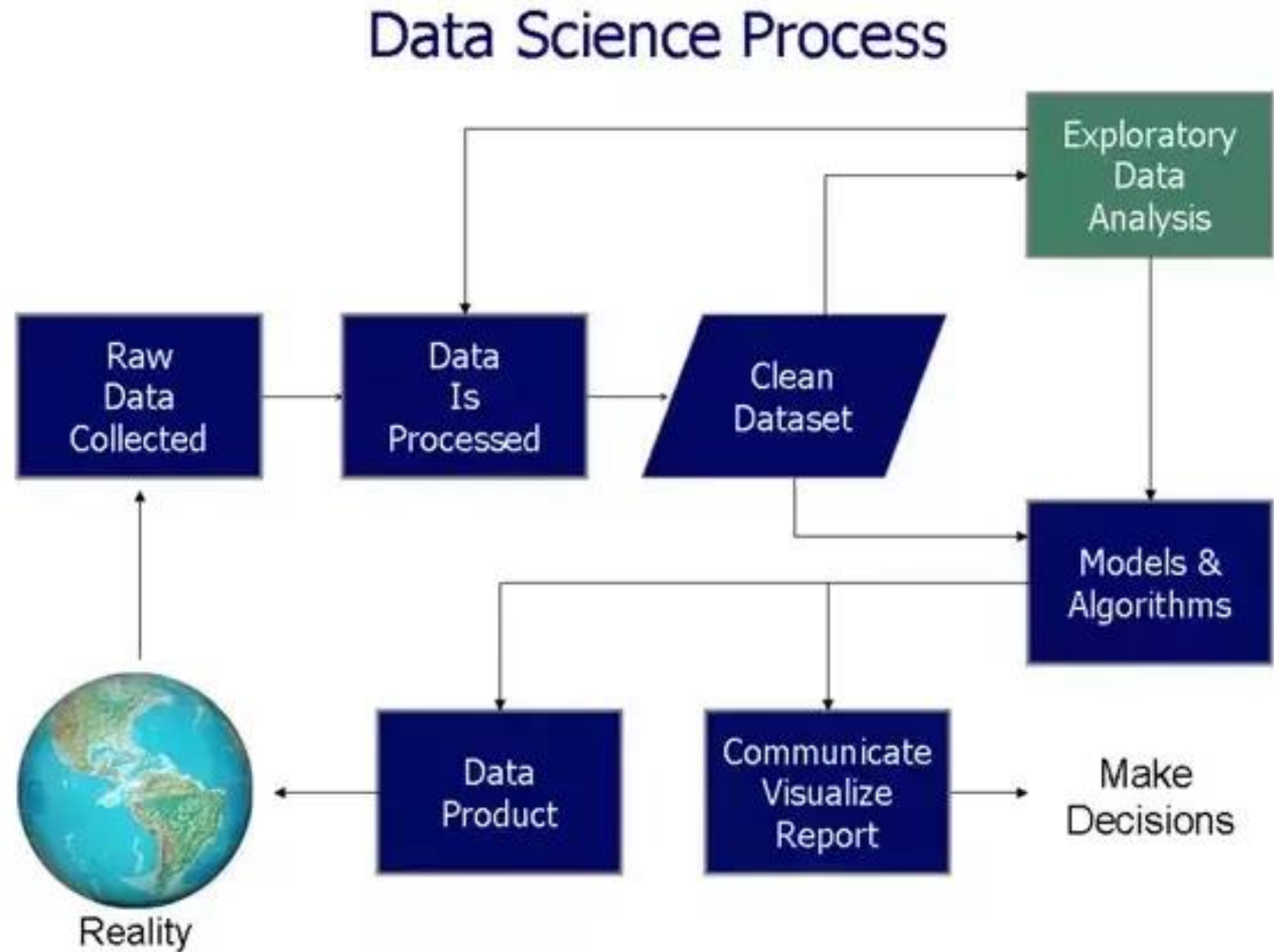
5. Data modeling

- Data mining techniques, machine learning, deep learning
- This is the most technical part

6. Visualization and communication

7. Deployment and maintenance of models

Involved Processes



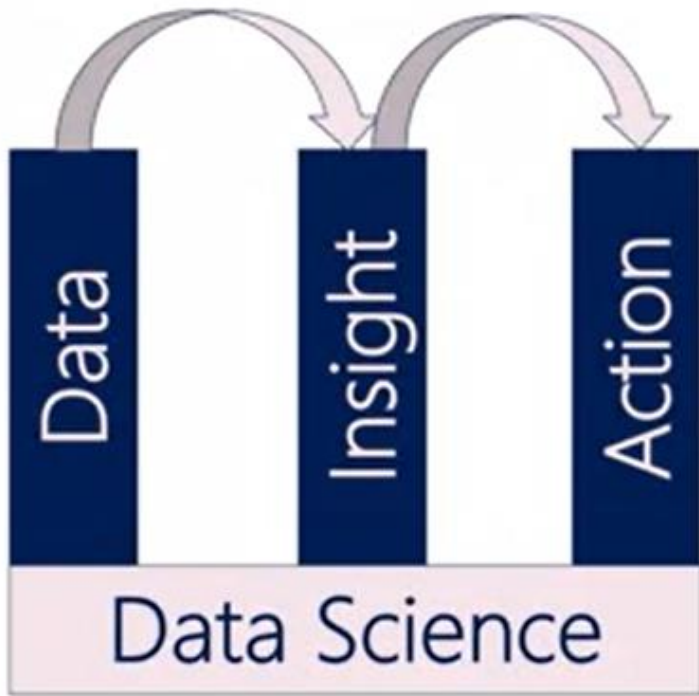
DS Team

- Data scientist (DS)
 - Prepares data, engineers features, most valuable skill: training models
- Data engineer (DE)
 - Data acquisition focus. Build data pipelines. Not uncommon to have 5:1 ratio DE:DS
- Data analyst (DA)
 - Assist DS with data preparation
- Application architect (AA)
 - Design complete solution; deploy and maintain models in production

Data engineering covers ...

- **Data Pipelines:** Building efficient and reliable systems to move data from various sources (databases, APIs, files, etc.) to data storage and processing platforms.
- **Data Storage:** Selecting and managing appropriate storage solutions (relational databases, data warehouses, data lakes, NoSQL databases) based on data characteristics and query patterns.
- **Data Transformation:** Cleaning, processing, and transforming raw data into a usable format for analysis.
- **Data Quality:** Ensuring data accuracy, completeness, consistency, and timeliness.
- **ETL (Extract, Transform, Load):** Designing and implementing processes to extract data from different sources, transform it, and load it into a target system.

Data product



“... a product that facilitates an end goal through the use of data”

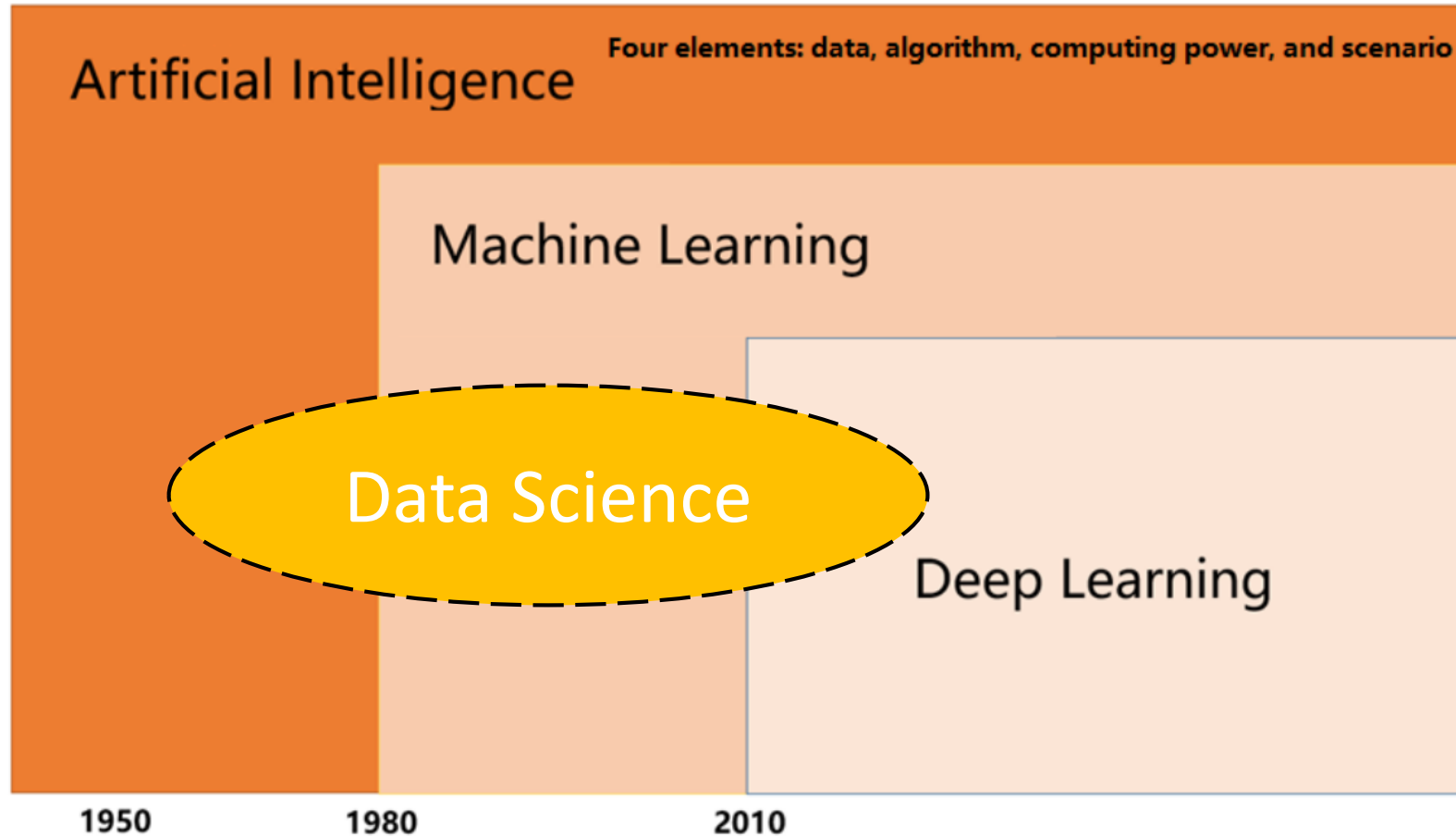
-- DJ Patil, Former U.S. Chief Data Scientist
in Data Jujitsu

Data product is the output of any data science activity. We generate actionable insights from big data using data science, e.g., recommender systems.

Data Products are
systems models
that help us to
understand data
in order to
gain insights and
make predictions

AI, ML, DL and DS

Image Credit: Huawei's HCIA-AI, 2022



AI → Enable machines to think; building machines that can mimic human intelligence

ML → Provide statistical tools to explore and analyze data (e.g., supervised, unsupervised, reinforcement learning (RL)); allowing computers to learn from data without being explicitly programmed

DL → Mimic how human brains work, to make machines learn (e.g., ANN, CNN, RNN, transfer learning (TL))

DS → Apply mathematical and statistical tools and methods/techniques from DL, ML and AI to data (big enough)

Machine Learning is ...

Machine learning is kind of artificial intelligence that is responsible for providing computers the ability to learn about newer data sets **without being programmed via an explicit source**.

- It focuses primarily on the development of several computer programs that can transform if and when exposed to newer sets of data.
- Machine learning and data mining follow the relatively same process but might not be the same.
- Machine learning follows the method of data analysis which is responsible for automating the model building in an analytical way.
- It uses algorithms that iteratively gain knowledge from data and in this process it lets computers find the apparently hidden insights without any help from an external program. In order to gain the best results from data mining, complex algorithms are paired with the right processes and tools.

DS vs. ML

Data Science

Explore many models, build and tune hybrids

Understand empirical properties of models

Develop/use tools that can handle massive datasets

Take action!



Machine Learning

Develop new (individual) models

Prove mathematical properties of models

Improve/validate on a few, relatively clean, small datasets

Publish a paper

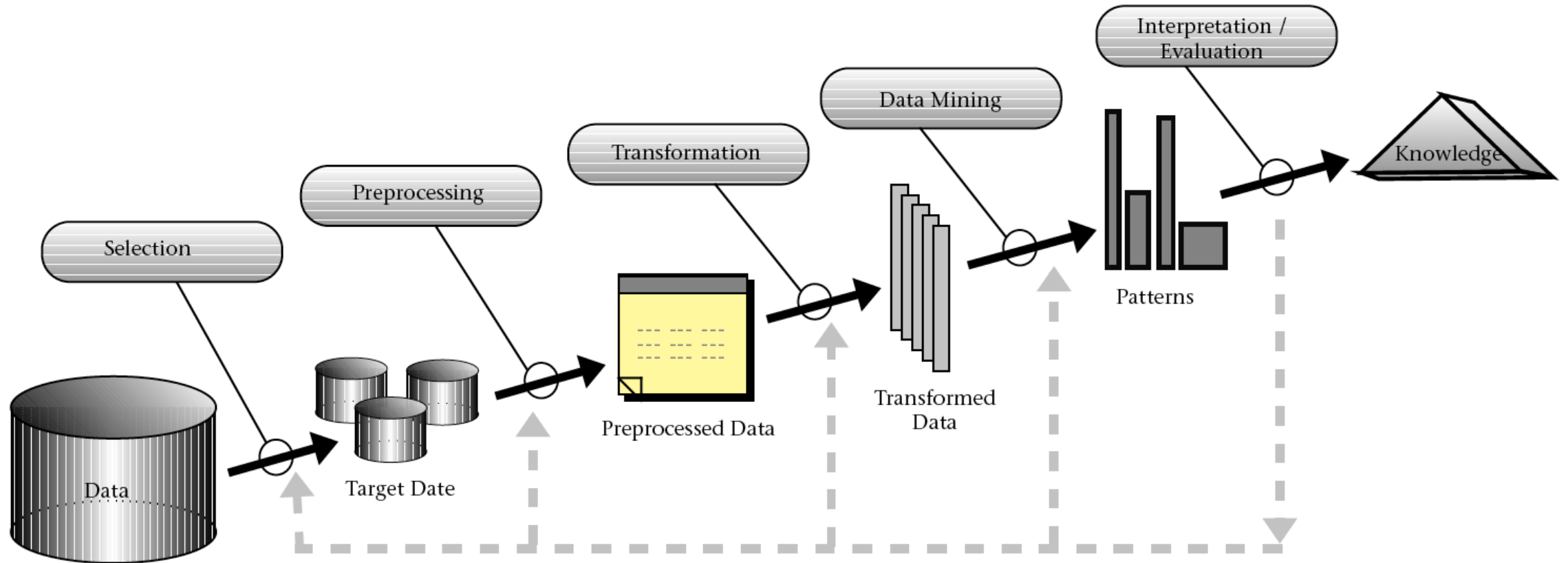
DS vs. AI



Data Science vs Artificial Intelligence

Factors	Data Science	Artificial Intelligence
Scope	Involves various underlying data operations	Limited to the implementation of ML algorithms
Type of Data	Structured and unstructured	Standardized in the form of embeddings and vectors
Tools	R, Python, SAS, SPSS, TensorFlow, Keras, Scikit-learn	Scikit-learn, Kaffe, PyTorch, TensorFlow, Shogun, Mahout
Applications	Advertising, Marketing, Internet Search Engines	Manufacturing, Automation, Robotics, Transport, Healthcare

Knowledge Discovery in Databases (KDD) → Data Mining (DM)



Data Mining (DM)

Data mining is simply the process of garnering information from huge databases that was previously incomprehensible and unknown and then using that information to make relevant business decisions.

- To put it more simply, data mining is a set of various methods that are used in the process of knowledge discovery (KD) for distinguishing the relationships and patterns that were previously unknown; it is a confluence of various other fields like artificial intelligence, virtual data room (VDR) management, pattern recognition, visualization of data, machine learning, statistical studies and so on.
- The primary goal of the process of data mining is to extract information from various sets of data in an attempt to transform it in proper and understandable structures for eventual use.
- Data mining is thus a process which is used by data scientists and machine learning enthusiasts to convert large sets of data into something more usable.

What will future wars be like? (V. Putin, 2017)



MARKETS

BUSINESS

INVESTING

TECH

POLITICS

CNBC TV

Putin: Leader in artificial intelligence will rule world

PUBLISHED MON, SEP 4 2017•2:33 AM EDT | UPDATED MON, SEP 4 2017•2:33 AM EDT

KEY POINTS

- Putin says that whoever reaches a breakthrough in developing artificial intelligence will come to dominate the world.
- Putin warned that “it would be strongly undesirable if someone wins a monopolist position”.

Dataset Repository

<https://archive.ics.uci.edu/>

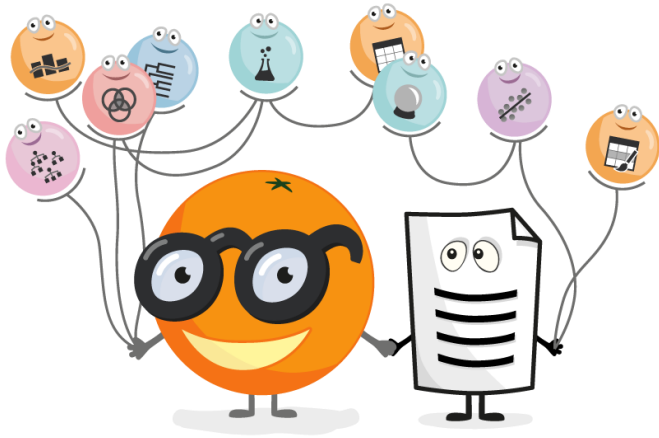


UC Irvine
Machine Learning
Repository

<https://www.kaggle.com/>

kaggle

Software Tools



<https://orangedatamining.com/>



RStudio Desktop

<https://posit.co/downloads>